

OPERATING SYSTEMS CONCEPT – LAB

EXP – 06: INTER-PROCESS COMMUNICATION USING SHARED MEMORY

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Activities Terminal May 14 20:26 snoordeen@localhost:~/snoordeen

snoordeen@localhost snoordeen$ cat writer.c
#include<stdio.h>
#include<sys/ipc.h>
#include<sys/shm.h>
#include<string.h>
int main(){
    key_t key = ftok("shmfile", 65);
    int shmid=shmget(key,1024,0666 | IPC_CREAT);
    char *str=(char*) shmatt(shmid, (void*)0,0);
    sprintf(str,"Hello");
    printf("Data written in memory: %s\n", str);
    shmdt(str);
    return 0;
}

snoordeen@localhost snoordeen$ cat reader.c
#include<stdio.h>
#include<sys/ipc.h>
#include<sys/shm.h>
int main(){
    key_t key = ftok("shmfile", 65);
    int shmid = shmget(key, 1024, 0666 | IPC_CREAT);
    char *str = (char*) shmatt(shmid, (void*)0,0);
    printf("Data read from memory: %s\n", str);
    shmdt(str);
    shmdt(shmid, IPC_RMID, NULL);
    return 0;
}

snoordeen@localhost snoordeen$ gcc writer.c -o writer
snoordeen@localhost snoordeen$ ./writer
Data written in memory: Hello
Data written in memory: Hello from the other side!
snoordeen@localhost snoordeen$ gcc reader.c -o reader
snoordeen@localhost snoordeen$ ./reader
Data read from memory: Hello
Data read from memory: Hello from the other side!
snoordeen@localhost snoordeen$
```